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REVIEW

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# Sustainable management of fall armyworm, *Spodoptera frugiperda* (J.E. Smith): challenges and proposed solutions from an African perspective

Shingirayi Nyamutukwa[a,b], Brighton M. Mvumi[a] and Peter Chinwada[c]

[a]Department of Agricultural and Biosystems Engineering, Faculty of Agriculture Environment and Food Systems, University of Zimbabwe, Harare, Zimbabwe; [b]Department of Research and Specialist Services, Plant Protection Research Institute, Harare, Zimbabwe; [c]International Institute of Tropical Agriculture, Southern Africa Research and Administration Hub (SARAH) Campus, Lusaka, Zambia

**ABSTRACT**
Fall armyworm (FAW), *Spodoptera frugiperda*, is threatening food security in Africa and thus sustainable management strategies are required. The pest has spread to over 44 countries in Africa since its first detection in 2016, causing maize yield losses valued at between US$2,531 and US$6,312 million per annum. Owing to FAW's damaging potential, many untested management strategies, including those of doubtful efficacy, are being used by smallholder farmers in Africa. We analysed existing and emerging FAW management strategies on the continent. Research and training has focussed on FAW identification, scouting, digital monitoring tools, pest distribution, natural enemy database, and FAW impact on crops. Gaps identified include lack of clear national policies and regulations, FAW identification challenges, absence of reliable and sustainable management options, and FAW insecticide resistance development. Conservation of FAW natural enemies could enhance sustainable natural control. Farmer Field Schools and mass rearing of natural enemies for augmentative release are sustainable FAW control strategies. The "push-pull" strategy in controlling FAW has potential in Africa. Existing policies and regulations to facilitate better FAW management are discussed.

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## 1. Introduction

Fall armyworm (FAW), *Spodoptera frugiperda* (J.E. Smith) (Lepidoptera: Noctuidae) is native to the tropical and subtropical regions of the Americas (Bateman et al. 2018; Devi 2018) and was first reported on the African continent in 2016 (Goergen et al. 2016). The pest has since spread to more than 44 countries in sub-Saharan Africa (SSA) (Feldmann et al. 2019), making it one of the most notorious cereal pests. Worldwide, at least 80 plant species have been reported as being FAW hosts, with maize, sorghum, rice and sugarcane being the most preferred (Pashley 1988; Pogue 2002; Casmuz et al. 2010; Day et al. 2017; Devi 2018; Montezano et al. 2018; Assefa and Ayalew 2019; Tefera et al. 2019).

There is a growing demand for food in Africa, especially maize, as the human population is increasing rapidly, necessitating the search and implementation of effective technologies for crop protection against insect pests, including FAW. In this paper, we critically review the evidence available on management of FAW, highlighting the existing gaps, sustainability, challenges, policy and regulatory issues that need to be addressed while considering contributions from regional and international efforts for the management of this pest. This information is critical in identifying and developing possible sustainable FAW management measures for use by agricultural practitioners while also guiding policy direction and resource-investment for effective management of the pest. This is the first review providing FAW management options, highlighting government and international efforts to manage the pest as well as focusing on policy issues to guide sustainable management of the pest in an African context. The review provides key policy points for consideration in reviewing pesticide registration, procurement of resources for FAW management and best management practices by farmers and other stakeholders in Africa.

## 2. Economic importance of FAW in crop production systems

Different authors have reported varying maize yield and economic losses due to FAW infestation in Africa (e.g. Day et al. 2017; Norenius 2018; FAO 2018a; Baudron et al. 2019; Chimweta et al. 2019;

**CONTACT** Brighton M. Mvumi mvumibm@agric.uz.ac.zw; mvumibm@hotmail.com Department of Agricultural and Biosystems Engineering, Faculty of Agriculture Environment and Food Systems, University of Zimbabwe, P.O. Box MP167, Mt Pleasant, Harare, Zimbabwe.

Kansiime et al. 2019; Kumela et al. 2019) (Table 1). Such variations in yield loss estimates within one country even during the same season or among different countries can be attributed to differences in time of planting and crop stage, geographical variations influencing climatic conditions, occurrence, diversity, and density of natural enemies, production practices such as intercropping, loss estimation methods, and conservation agriculture and farmer practices in FAW control.

Research on maize yield losses by FAW has been conducted in a number of African countries (Table 1), but no studies exist on the impact of the pest on other cereal crop hosts, such as sorghum and millet that ensure food security in semi-arid areas due to their drought tolerance. Studies by Koffi et al. (2020) in Togo only focused on FAW infestation levels from 2016 to 2018 (ranging from 15 to 71%) without giving actual yield losses due to the pest. Information on yield losses is critical in quantifying the value of maize loss due to FAW and facilitates comparisons of pest impact among different countries.

Different economic loss values due to FAW attack have been reported by various authors. Day et al. (2017) estimated annual losses between US$2,531 and US$6,312 million in 12 African countries (Table 2). Using information on economic losses from their studies in Ghana (US$177 million) and Zambia (US$159 million), Rwomushana et al. (2018) estimated economic losses between US$1.1 billion and US$4.7 billion in the 12 African countries. Considering this increased economic loss in the 12 countries, it is necessary to collect statistics from all affected countries to generate country-specific losses and come up with a more accurate estimate of the continental economic loss.

Due to its adverse effect on yields, FAW has the potential to negatively affect the availability of maize seed for coming seasons (Prasanna et al. 2018). Because of its polyphagous feeding habit, it is difficult to manage FAW as it can survive and reproduce on alternative hosts when its main cereal hosts are off-season. For example, in a comprehensive systematic study, FAW larvae were recorded on 353 host plant species belonging to 76 plant families, mainly Poaceae (106), Asteraceae (31), and Fabaceae (31) (Montezano et al. 2018). Chimweta et al. (2019) reported the presence of FAW on pigweed (*Amaranthus* spp.) and jungle rice (*Echinocloa colona* L.) before maize planting. Itch grass [*Rottboellia cochinchinensis* (Figure 1A), Lour (Clayton)], brachiaria grass (*Brachiaria* spp.) (Figure 1B) and rapoko grass [*Eleusine indica* L. (Gaertn)] are examples of common weed species on which FAW larvae feed alongside maize plants in fields (Nyamutukwa, personal observation). Thus, understanding the ecology of FAW, including its full host range, is important in the development of Integrated Pest Management (IPM) programmes, especially when destruction of alternative hosts is necessary to minimise pest carryover from one season to another.

## 3. Co-occurrence and co-existence of FAW and other lepidopteran pests

Before the arrival of FAW on the continent (Goergen et al. 2016), notable lepidopteran pests that had been foraging on maize included stem borers [e.g. *Busseola fusca* Fuller, *Chilo partellus* (Swinhoe), *Sesamia calamistis* (Hampson), and bollworm (*Helicoverpa armigera* Hübner)] (Zhou and Overholt 2001; Chinwada et al. 2003, 2014; Prasanna et al. 2018; FAO and CABI 2019). Indirect identification of FAW through plant damage symptoms has brought about challenges to farmers and agricultural extension workers due to similarities with damage caused by these other lepidopteran species. Similar damage symptoms include pinholes, windowing, shot holes, cob damage, tassel and silk damage. Correct identification of the pest is critical for correct choice and application of control measures. With many farmers now recognising FAW as the most important cereal pest overtaking stem borers (Kumela et al.

**Table 1.** Maize yield losses due to FAW in some sub-Saharan African countries.

| Country | Estimated yield loss (%) | Year or season | Source |
|---|---|---|---|
| Benin | 49 | 2018/19 | Houngbo et al. (2020) |
| Ethiopia | 32 | 2017 | Kumela et al. (2019) |
| Ghana | 26.6 | 2018 | Rwomushana et al. (2018) |
| Kenya | 47.3 | 2017 | Kumela et al. (2019) |
| | 53–54 | 2017 | De Groote et al. (2020) |
| | 42 | 2018 | De Groote et al. (2020) |
| Mozambique | 65 | 2018 | Norenius (2018) |
| Namibia | 13 | 2017 | FAO (2018a) |
| Zambia | 28 | 2016/17 | Kansiime et al. (2019) |
| | 35 | 2018 | Rwomushana et al. (2018) |
| Zimbabwe | 11.6 | 2017/18 | Baudron et al. (2019) |
| | 58 | 2017 | Chimweta et al. (2019) |

**Table 2.** Estimated lower and upper yield and economic losses due to FAW in 12 maize-producing African countries.

| Country | Maize production (3-year mean) (million tonnes) | Value of maize with no FAW (3-year average FAO stats) [US$ million] | Yield loss (lower) (million tonnes) | Yield loss (upper) (million tonnes) | Mean yield loss (million tonnes) | Economic loss (lower) [US$ million] | Economic loss (upper) [US$ million] |
|---|---|---|---|---|---|---|---|
| Benin | 1.2853 | 376.5 | 0.2956 | 0.7358 | 0.5304 | 86.6 | 215.6 |
| Cameroon | 1.6657 | 697.8 | 0.3192 | 0.7944 | 0.6874 | 133.7 | 332.8 |
| Democratic Republic of Congo | 1.1734 | 343.7 | 0.2545 | 0.6334 | 0.4842 | 74.5 | 185.5 |
| Ethiopia | 6.6283 | 1,580.2 | 1.2272 | 3.0547 | 2.7352 | 292.6 | 728.3 |
| Ghana | 1.8255 | 629.8 | 0.4016 | 1.2139 | 0.8243 | 138.5 | 418.8 |
| Malawi | 3.3449 | 979.7 | 0.7693 | 1.915 | 1.3803 | 225.3 | 561.0 |
| Mozambique | 1.2472 | 365.3 | 0.2862 | 0.7124 | 0.5147 | 83.8 | 208.7 |
| Nigeria | 9.3027 | 3,271.8 | 2.1291 | 5.2997 | 3.8389 | 748.7 | 1,863.6 |
| Uganda | 2.7483 | 805.0 | 0.5589 | 1.3911 | 1.1341 | 163.7 | 407.5 |
| Tanzania | 5.7326 | 1,679.1 | 1.3013 | 3.239 | 2.3656 | 381.2 | 948.8 |
| Zambia | 2,913.0 | 500.9 | 0.7281 | 1.4561 | 1.1540 | 125.2 | 250.4 |
| Zimbabwe | 1.1041 | 360.7 | 0.2348 | 0.5844 | 0.4556 | 76.7 | 190.9 |
| **Total** | **38.971** | **11,591** | **8.506** | **21.030** | **16.105** | **2,531** | **6,312** |

Source: Day et al. (2017)


**Figure 1.** FAW damage on itch grass, *Rottboellia cochinchinensis* (A) and brachiaria grass, *Brachiaria* spp. (B). (Photographs by S. Nyamutukwa).

2019), this shift requires better understanding of the behaviour of FAW for effective management. For example during winter, stem borer larvae hibernate in crop residues while FAW does not hibernate (Hailu et al. 2018; Nagoshi et al. 2019) and therefore will need to be managed even on crops grown off-season. However, some control strategies may target both cereal stem borers and FAW if they are found feeding on the same host depending on the plant part affected. Despite observing both FAW and stem borer larvae feeding on the same maize plant (Chinwada, personal observation; Nyamutukwa, personal observation), actual predation of larvae of the latter by FAW larvae has not been noted or reported. This necessitates in-depth studies to unpack the interactions between the two pest species. Elsewhere, there have been several reports of FAW cannibalism (Raffa 1987; Chapman et al. 1999a, 1999b) which require validation in the SSA context.

## 4. Influence of climate on spreading, establishment, and multiplication of FAW

Variations in climatic factors are known to be critical in influencing pest species spectrum, abundance, population dynamics, and action thresholds in all IPM strategies (Blanco et al. 2014). Apart from host availability (Montezano et al. 2018), temperature and precipitation are the most influential factors influencing the development and spreading of FAW (Early et al. 2018; Wang et al. 2020) and their natural enemies (López et al. 2018). In SSA, climate is generally favourable to FAW spread (Rangi and Vos 2017; FAO and CIMMYT 2018; Prasanna et al. 2018; FAO and CABI 2019). Recent studies by Nurzannah et al. (2020) showed an increase in area under FAW attack at monthly temperatures and rainfall of at least 19.1 °C and 14.3 mm, respectively. The peak period of FAW attack occurred when there was abundant host availability due to high rainfall received and favourable temperature. Wang et al. (2020) also gave a close temperature range of between 19.2 °C and 29.7 °C as the most suitable for FAW development.

The potential for geographical distribution of invasive organisms can be obtained using predictive models. Rangi and Vos (2017), using prediction models, established that FAW had the potential to invade Madagascar and pest presence was later confirmed by reports (Prasanna et al. 2018; FAO 2018b, 2019). Prediction models by Day et al. (2017) may need to take into account irrigation programmes established in areas perceived as dry and unsuitable for pest establishment thus improving accuracy of the modelling.

Temperature is an important factor in insect-host-pathogen interactions, and plays a key role in transmission of entomopathogens (Elderd and Reilly 2014). In their study, López et al. (2018) suggested that annual precipitation of less than 176 mm and temperatures between 15.8 °C and 32.5 °C influenced the establishment and viability of entomopathogens that regulate FAW. At temperatures below 17 °C, Wang et al. (1997) reported reduced egg parasitism of the European corn borer, *Ostrinia nubilalis* Hübner by *Trichogramma ostriniae* Pang & Chen. This information is critical for survival of parasitoids, as *Trichogramma* spp. have been reported parasitizing FAW as well (Prasanna et al. 2018). Fonseca-Medrano et al. (2019) reported lower populations of three lepidopteran pests, including FAW, when precipitation, temperature and humidity were at their highest. FAW larval and pupal survival in plant whorls and in the soil is affected by precipitation due to drowning of moths in funnels and collapsing of pupation tunnels (Early et al. 2018). This information should guide farmers in their decisions to spray against FAW when heavy rains have been experienced or are likely to fall.

## 5. Regional and national approaches in management of FAW – gaps and challenges

### 5.1. Chemical control

Most farmers rely on chemical control as the first line of defence against FAW (Kumela et al. 2019), but the strategy comes with high costs, and environmental and human health risks (Yigezu and Wakgari 2020). Many smallholder farmers use knapsack sprayers as the most common equipment in FAW control (Day et al. 2017; Kumela et al. 2019). A review by Bateman et al. (2018) on active ingredients registered for use against FAW indicates that spinosad is the widely used biopesticide in SSA. However, this must have been based on a very small sample size as by then many FAW plant protection products were still under evaluation on the continent. If a survey was to be conducted now, it is likely that the most common active ingredient in FAW plant protection products would be emamectin benzoate (Chinwada, personal observation). In 2016, Zambian and Ghanaian governments offered free pesticides to farmers (Rwomushana et al. 2018) and, likewise, the Zimbabwean Government, in partnership with FAO, procured and distributed chemicals and knapsack sprayers to affected farmers. The pesticides distributed include carbaryl 85% WP, emamectin benzoate, and acetamiprid.

Maize production in Africa is predominantly by smallholder farmers, while in the Americas commercial farmers dominate. Commercial farmers in USA, Brazil, and Argentina are largely dependent on genetically modified (GM) maize and use technologies unavailable to smallholder farmers (Hruska 2019). Under smallholder production, application of chemicals is time-consuming and effectiveness is compromised by inaccurate dosages and time being shared with other field operations such as weeding compared to mechanised crop protection in commercial production systems. Inaccurate dosages may result in poor control of FAW or trigger insecticide resistance development and hence creating more challenges in controlling the pest.

The systemic seed insecticide Fortenza™ Duo was proactively deployed in Zambia and Zimbabwe at the end of 2018 while trials were still taking place (Chinwada, personal communication). Fortenza™ Duo is available as a twin-pack of the commercial insecticidal products Fortenza® 600 FS (cyantraniliprole 600 g/L and and Cruiser® 600 FS (thiamethoxam 600 g/L). The recommended mixing rate for maize seed treatment is 292 mL of each insecticide per 100 kg of maize seed, giving 5.84 L of the mixture per tonne of seed. Mixing and treatment are done by seed companies, so that farmers purchase a seed-packaged pest management technology. To enable differentiation between Fortenza™ Duo-treated and non-treated seed, packs containing treated seed have a sticker on them. According to the Fortenza™ Duo information pack, the seed treatment potentially offers protection against FAW for the first 3–4 weeks of growth post-emergence. While cyantraniliprole affects FAW larvae through stomach action (mainly) and contact effect, thiamethoxam offers protection against a range of root feeders and sap suckers thus enabling better establishment of the seedlings and optimized uptake of water and nutrients from the soil before being attacked by the pest (Syngenta 2018).

Generally, chemicals kill FAW larvae on contact or through ingestion. Most chemicals used in FAW control are effective when applied correctly, at the right dosages, and at the right time, but require that the person applying the chemicals use personal protective equipment (PPE). As use of PPEs is a huge challenge for most smallholder farmers (Prasanna et al. 2018) this poses human health risks. It is very difficult to give a rate of application per hectare to a farmer, as this parameter depends on crop growth stage, the speed at which the applicator moves, and the infestation levels or pest distribution in the field. Smallholder farmers also find it difficult to measure land area and even the right quantities of the pesticide to be applied, mainly associated with illiteracy (Oben et al. 2015), lack of awareness or lack of diligence. Rather, it has been easier to provide the amount of chemical to be diluted per knapsack sprayer. Chemical applications using knapsack sprayers also come with challenges of drift and inhalation risks when the lance is raised in response to changes in crop height as plants grow. In addition, there are challenges with the period that farmers should wait until it is safe to re-enter the field – the Restricted Entry Interval (REI) – for scouting pests or assessing effectiveness of a previous spray. Challenges with pre-harvest interval when chemicals are applied to maize at the soft dough stage are not fully addressed, especially when maize cobs are harvested for consumption as green mealies (cobs harvested at soft dough stage for consumption after being roasted or boiled). Maize is a staple crop in most African countries (Midega et al. 2015; Kumela et al. 2019). Because green mealies are a major source of income, especially in urban areas, it is very difficult to recommend abandonment of green mealies production (FAO and CIMMYT 2018) at any particular time of the year for purposes of managing FAW. The recommendation may also not yield any tangible benefits due to the pest's wide host range.

Issues of environmental concerns have also been reported previously, especially the effect of chemicals on bees, natural enemies, and antagonistic fungi, either as direct sprays in fields or as non-targets in pheromone traps (Malo et al. 1999, 2004; FAO and CIMMYT 2018; Kumela et al. 2019). Although maize is a wind-pollinated plant that does not require physical pollination, bees forage for pollen on tassels and some have also been caught in traps of different colours (Wheelock 2014). Trap colour and insecticides used in FAW traps are probable contributory factors in bee mortalities. This consequence necessitates the use of "softer" chemicals or non-chemical methods. The Pesticide Action Network International currently lists 118 chemicals as being highly toxic to bees out of 338 including, among others, acephate and carbaryl that are used for FAW control (PAN International 2021). In Africa, a number of countries have banned the use of highly hazardous pesticides (HHP) according to the PAN International list with Zimbabwe banning 21, Zambia 3, South Africa 16, Kenya 14, Ghana 9, and Malawi 14. However, these numbers are lower when compared to Brazil (banned 131) and the European Union (banned 175). Examples of banned chemicals on the list include endosulfan, lindane, mercury compounds, trichlorfon, and methamidophos. However, in smallholder African farming systems, banning HHPs from use against FAW does not necessarily lead to their unavailability on the market, especially in cases where such farmers also grow cotton and tobacco where these chemicals may still be approved for use. Since this classification of chemicals as HHPs or non-HHPs may result in several chemicals that could be used in FAW management, being banned now or in the near future, there is need to first put in place a regionally harmonised FAW IPM approach composed of several management options.

The use of synthetic insecticides has also brought challenges of FAW resistance development (e.g. Young and McMillian 1979; Yu 1991) although in Africa evidence of such is still scanty because the pest only occurred recently on the continent (Goergen et al. 2016). Considering that strains of FAW in Africa and Asia are geographically separated from those in the Americas (Nagoshi et al. 2017, 2019, 2020), the inheritance of such traits should be considered for effective pest management. The use of herbicides such as acetochlor, atrazine and glyphosate by farmers in Zambia (Kansiime et al. 2019) requires further research on their effectiveness in indirect FAW control (i.e. through controlling host weeds which FAW larvae can feed on while crawling on the ground or when the main crop is off-season). While Tambo et al. (2020) attributed farmers' erroneous use of fungicides in controlling FAW to limited knowledge of the pest, it is also important to provide information on types of chemicals and pests they control in order to promote the appropriate and judicious use of pesticides. Insecticides reduce abundance of natural enemies of pests (Harris-Shultz et al. 2015) and information on the insecticides' impact on natural enemy populations on the African continent is scanty. Zhu et al. (2015) reported that FAW strains resistant to Bt toxins expressed in transgenic crops were also resistant to organophosphates, which could be independent events. This information is critical in the development of effective IPM strategies when such evidence of resistance across strategies is reported. In addition, the cessation of insecticide regimes allows natural enemies to take over FAW population regulation (Wightman 2018), especially when there is limited research-guided insecticide rotation regimes for adoption by farmers across the continent.

### 5.2. Cultural control

Different cultural practices have been reported to be effective in reducing FAW damage in maize, such as frequent weeding and minimum or zero tillage (Baudron et al. 2019), avoiding late planting of maize (Assefa and Ayalew 2019; Kansiime et al. 2019), intercropping with appropriate crops, but avoiding maize-pumpkin intercrops, as maize in such intercrops showed increased FAW damage (Baudron et al. 2019; Harrison et al. 2019). Handpicking, ash application in plant whorls and crushing of FAW eggs and caterpillars have been reported to reduce FAW damage (Assefa and Ayalew 2019; Hruska 2019). However, because they are labour-intensive, practices such as handpicking and crushing of eggs and caterpillars are only practical for small-cropped areas. The main challenge with some cultural control strategies is that while they aim at reducing FAW damage, in fact some of them do not encourage natural enemy population build up. For example, while crushing eggs or burying pupae through ploughing under reduces FAW numbers, biocontrol agents such as parasitoids rely on the same as hosts. The simple question that arises is: Were the eggs that were crushed parasitized or not? If the crushed eggs were going to yield parasitoids, this renders the strategy incompatible with IPM, as it does not encourage population build-up of biological control agents. It is very difficult to identify a parasitized FAW host and this complicates FAW management through conservation of parasitoids. Since no studies to determine potential FAW parasitoid population reductions from such practices have been conducted anywhere, the likely adverse impacts of such mechanical control methods remain unclear. This presents an opportunity for researchers to make some cultural control practices as compatible as possible in an IPM approach; thus further testing is required before adoption. A possibility for removing FAW eggs from fields coupled with rearing may provide information on the extent of regulation by parasitoids under Farmer Field Schools (FFS).

### 5.3. Genetically modified plants

In the tropical and subtropical regions of the Americas, FAW is predominantly managed through the use of genetically modified (GM) crops (Hruska 2019). A number of GM crops have been introduced on the African continent for FAW control. For example, Bt maize varieties were introduced in Kenya where they demonstrated partial control of FAW (Norenius 2018). Upon its introduction, the GM maize was expected to tolerate stem borers known to reduce maize grain yield by an average of 13% or 400,000 tonnes of grain, equivalent to the normal yearly amount of maize that Kenya imports (ISAAA 2016a). Ironically, these introductions of Bt maize varieties are being made at a time when field-evolved resistance development to such transgenic varieties by FAW has already been reported in the Americas (e.g. Bernardi et al. 2015; Yang et al. 2016; Tabashnik and Carrière 2017; Lourenço et al. 2018). Globally, there are reports of field-evolved resistance to Bt proteins such as Cry1F and Cry1Ab expressed in maize transgenic events (Farias et al. 2014, 2016; Omoto et al. 2016; Boaventura et al. 2020). This compromises the sustainability of such an approach considering that the strains of FAW introduced in Africa originated from the same source (Botha et al. 2019).

Variations in national policies of African countries on acceptability of genetic modification of crops present a big challenge if the strategy is to be regionally adopted. Carzoli et al. (2018) identified research, data provision and education as key components to policy change. Challenges with introductions of GM crop varieties on the market are dependent on political will to approve such releases with delays in passing biosafety bills being experienced (Norenius 2018). On the other hand, sustainability of GM crops is threatened by development of resistance by FAW to Cry1Ab protein as reported in South Africa (Botha et al. 2019). In their study, Botha et al. (2019) concluded that the moderate susceptibility of FAW to Cry1Ab (4–35%) could be due to populations arriving on the African continent already possessing alleles with resistance to this protein. Gómez et al. (2018) showed that maize expressing Cry1A toxin does not effectively control FAW but controls other lepidopteran pests while FAW is controlled by maize expressing the Cry1Fa toxin. The development of transgenic crops targeting FAW should therefore utilise those strains producing effective toxins against the pest.

Lack of rigorous scientific data and analysis have contributed to failure by governments to pass bills on biosafety (Norenius 2018). While FAO has coordination role to ensure regional harmonised approaches to FAW management are developed, its capacity falls short when it comes to national policy formulation and decision-making on genetically modified organisms (GMOs) as this is a prerogative of member states. However, FAO could provide legal and technical advice to governments in areas of national biotechnology strategies and formulation of biosecurity frameworks. While coordinating regional approaches for effective FAW management, it is considered too early to draw conclusions on potential use of GMOs in FAW control (FAO and CIMMYT 2018).

The development of GM crops against lepidopteran pests has resulted in questions being raised on unintended effects of the transgenic crops in the surrounding ecosystems such as effect of Bt toxins on non-target lepidopterans as well as earthworms in soil (Eckerstorfer et al. 2007). Zhong et al. (2000) discovered that one Cry1Bb1 crystalline protein from Bt strain YBT-226 was toxic to three insect orders – Diptera, Coleoptera, and Lepidoptera. The authors reported mortality at low to moderate concentrations to larvae of the house fly (*Musca domestica*, Diptera), cottonwood leaf beetle (*Chrysomela scripta*, Coleoptera), and tobacco hornworm (*Manduca sexta*, Lepidoptera) providing first evidence of broad-spectrum toxicity.

The development of resistance by FAW to Bt toxins in Brazil has been documented in just three years of the first release of Bt maize (Omoto et al. 2016; Fatoretto et al. 2017). Botha et al. (2019) and Visser et al. (2020) reported evolution of resistance by FAW and stem borer populations to Bt maize in South Africa. The development of insect resistance in the three major commercial Bt crops – maize, cotton and soyabean that are attacked by FAW and other lepidopterans pests (Pashley 1988; Casmuz et al. 2010; Yang et al. 2016) – already presents a challenge when other countries consider adoption of the technology. This represents a setback to management of lepidopterous maize pests by the technology as resistance to Bt maize by the stem borer, *B. fusca* had been reported earlier (van Rensburg 2007). Yang et al. (2016) reported development of cross-crop resistance by a FAW population selected with Cry1F against three single genes and pyramided Bt cotton products, threatening the sustainability of GM crops in FAW management. The management of cross-resistance between single gene and pyramided Bt cotton products requires careful selection of the resistance traits (Niu et al. 2013).

Although ways of managing insect pest resistance have been suggested (Vilarinho et al. 2011; Murúa et al. 2019; Visser et al. 2020), information from countries growing these GM crops on the frequency of exposure to Bt toxins and its impact on resistance development is scarce. The culture of saving seed to plant in the next season practised by 98% of smallholder farmers in Africa as well as the high cost of transgenic maize seed, among other factors, threaten the adoption and sustainable use of the technology (FAO 2018c). While there are benefits of growing GM crops (Carzoli et al. 2018), there is need to address challenges related to their adoption by investing in human capacity development and biosafety knowledge-sharing through massive training programmes, especially targeting policy-makers. Addressing issues of insect resistance management like refuge compliance (Fatoretto et al. 2017) may promote the adoption of the technology so that developing countries benefit through improved crop yields and reduced production challenges (Carzoli et al. 2018).

### 5.4. Push-pull strategies

The “push-pull” strategy relies on planting repellent crops or plants that are more attractive to ovipositing FAW moths than the main crops. Research in Kenya, Uganda and Tanzania has shown that the strategy significantly reduces FAW infestations and subsequently yield losses in a maize crop (Midega et al. 2018; FAO 2018b). The existence of both FAW and stem borers in cereal production systems can be well-managed through “push-pull” strategies (Hailu et al. 2018).

In Zimbabwe, promising FAW management results are being obtained from on-going “push-pull” trials using brachiaria grass and maize-legume intercrops (Baudron et al. 2019). Seed companies are among some of the stakeholders involved in “push-pull” strategies for controlling FAW in Zimbabwe. However, the strategy is mainly applicable to smallholder farming communities (FAO 2018b) with limited evidence of its applicability under large-scale production systems, possibly due to the mechanisation requirements in the farming sector. The limited availability of grass seed used in the “push-pull” strategy may also compromise quick adoption of the strategy. There are efforts to sell seed as a pack comprising maize and grass seeds. The use of brachiaria grass as stock-feed may promote its adoption and use in “push-pull” strategies. In general, however, the success with “push-pull” in East Africa does not necessarily mean that this is replicable in Southern Africa where land is not a limiting factor and maize is not planted on the same field every season.

### 5.5. Biological control

Most lepidopteran pests in cropping systems have natural enemies regulating their populations. While several indigenous natural enemies regulating FAW populations in Africa are being documented before classical biological control options are explored (Sisay et al. 2018, 2019; Kenis et al. 2019; Agboyi et al. 2020), very little has been done to study the compatibility of different management options available to farmers. In this context, biological control of FAW should not be considered in isolation, but as part of an ecosystem. Some strategies used in managing target pests often ignore the important role played by the associated natural enemies and hence are not compatible when included in an IPM context. As previously mentioned, the manual crushing of FAW eggs or larvae without considering its impact on egg and larval parasitoids could lead to reduction in the populations of natural enemies within the two guilds. It can, however, also be argued that those strategies deemed to be incompatible with biological control are still needed at certain stages of crop growth, especially those stages where natural enemy populations are still very low.

In nature, biological control agents may attack several lepidopteran pests (Southon et al. 2019). Literature shows that natural enemies regulating FAW have been in existence before the advent of the pest in Africa (López et al. 2018; Sisay et al. 2018; Agboyi et al. 2019; Kenis et al. 2019; Southon et al. 2019). This information, therefore, calls for strategies that consider alternative hosts to conserve these natural enemies.

Natural enemies documented as attacking FAW as well as stem borers, African armyworm and bollworms include nucleopolyhedroviruses, *Bacillus thuringiensis*, and fungal entomopathogens [e.g. *Beauveria bassiana* (Balsamo) Vuillemin, *Metarhizium anisopliae* (Metch.) Sorok.] (Akutse et al. 2019; Ngangambe and Mwatawala 2020). Parasitoids play an important role in reducing crop damage caused by insect herbivores (de Lange et al. 2018). Several authors have reported recoveries of the egg parasitoid *Telenomus remus* Nixon (Hymenoptera: Platygastridae) from FAW and other lepidopterans (Ndemah et al. 2001; De Freitas et al. 2014; Kenis et al. 2019; Liao et al. 2019; Sisay et al. 2019; Agboyi et al. 2020). More than 50% larval parasitism by *Cotesia icipe* (Fernández-Triana & Fiaboe) has been reported under laboratory conditions (Prasanna et al. 2018). However, from surveys carried out in Ethiopia, parasitism by *C. icipe* ranged from 33.8 to 45.3% (Sisay et al. 2018). About 30.6% egg parasitism by *T. remus* was reported on field-collected FAW egg masses (Liao et al. 2019). Assefa and Ayalew (2019) reported FAW parasitism by different larval parasitoids ranging from 4.6 to 45.3%.

Previous work indicating the yield benefits due to natural enemies controlling stem borers is documented (Ndemah et al. 2003; Kebede et al. 2018). There is potential for such benefits to be also realised on FAW as *Telenomus* spp. are already present in Africa (Kenis et al. 2019). Augmentive releases of *T. remus* can achieve more than 90% parasitism (Cave 2000). The ability of a parasitoid to exploit several hosts is critical for its survival, while augmentative releases boost control of the target pests in cropping environments. With the presence of *T. remus* and other parasitoids already confirmed on the continent (Assefa and Ayalew 2019), prospects for mass rearing, redistribution and release can be explored. The genus *Cotesia* (Hymenoptera: Braconidae) contains several species that are known to regulate populations of stem borers (Chinwada et al. 2008) as well as of FAW (Sisay et al. 2019; Agboyi et al. 2020). Recoveries of the stem borer tachinid parasitoid *Sturmiopsis parasitica* (Curran) (Chinwada et al. 2014) are being made on FAW in Zimbabwe (Nyamutukwa, personal observation). The egg-larval parasitoids *Chelonus* spp. (Hymenoptera: Braconidae) have also been recovered from FAW and stem borers in several studies (Wyckhuys and O'Neil 2006; Ordóñez-García et al. 2015; Hay-Roe et al. 2016; Capinera 2017; Sisay et al. 2018; Hruska 2019). Probably FAW natural enemies that have been recorded to date in SSA are only a fraction of the real number that will ultimately be found to be utilizing the pest as a host on the continent. Thus, the more natural enemy species are identified, the greater are the options for harnessing biological control in FAW IPM programmes. In general, the existence of common or similar natural enemies in many African countries (Kenis et al. 2019; Agboyi et al. 2020) provides a platform for harmonised area-wide biocontrol of FAW or redistributions of effective ones. However, biological control offers a sustainable way of managing FAW when combined with other human health and environmentally friendly approaches such as use of "soft" chemicals, scouting and handpicking in an IPM programme.

In the case of tephritid fruit-flies, Klungness et al. (2005) and Jang et al. (2007) developed the augmentorium technique to sequester fruit fly adults while releasing their natural enemies back into the environment. This method was reported to be more effective in controlling fruit flies when compared to burying infested fruit 0.46 cm underground. It is against this backdrop that the Lepi-Augmentorium (LA) (Figure 2) is proposed and adapted as a new concept to ensure that natural enemies of FAW are released back into the environment to continue suppressing the pest in an IPM approach. The LA uses a container such as a cage into which all lepidopteran life stages (eggs, larvae and pupae) collected from the field during routine scouting are

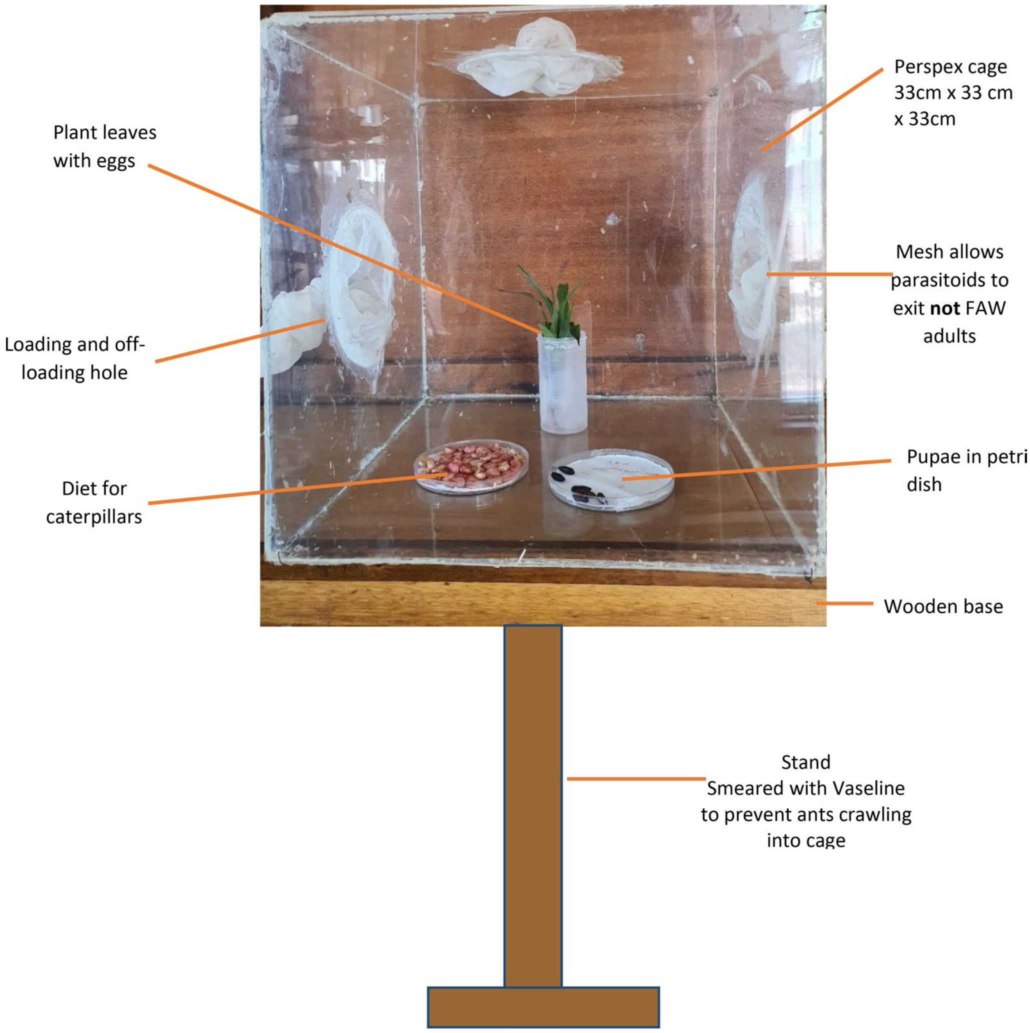


**Figure 2.** The Lepi-Augmentorium for releasing emerged parasitoid adults back into the environment [Concept adapted from Klungness et al. (2005)].

placed and reared. Adults are excluded from these collections, as they are free-living. Eggs, larvae and pupae are critical for the survival and conservation of natural enemies. The LA has all sides covered with perspex or glass material but has access holes covered by mesh cloth on its sides (two or three) for loading and off-loading samples. The mesh cloth does not allow lepidopteran moths to escape from the cage and are useful in ventilation.

The LA should be mounted under a shade in such a way that it is supported above ground (at least 1.5 m) for easy access by workers. The cage should have a roof placed above it for protection from rains that may fill it up with water. Soil is placed at the base of the container to provide a medium for last-instar FAW larvae to burrow into and pupate. Handpicking of FAW larvae is one of the control strategies mainly practised by smallholder farmers (Rwomushana et al. 2018; Kumela et al. 2019) thereby forming the basis for placement of field-collected FAW life stages (eggs and larvae) in the LA for possible release of any emerging adult parasitoids into the environment rather than crushing the collected pest life stages. Rearing of insects has been mainly for research purposes (Polaszek 1999) but this LA approach incorporates civil society as suggested by Ruzzeddu et al. (2018) in their training for the success of the biological control programme. This programme will help farmers identify natural enemies of FAW for future conservation and decision-making should there be any insecticide-spraying programme. The strategy also contributes to the establishment of introduced natural enemies.

The existence of FFS in over 90 countries (FAO 2018b) presents the best platform to promote conservation of FAW natural enemies by training farmers and agricultural extension staff through discovery-learning on the establishment of such LA-augmentoria and the identification of existing parasitoids regulating FAW. Diet for FAW larvae placed in the LA can be sprout leaf whorls, maize or bean sprouts (Raffa 1987). Research institutes of crop protection may also embrace mass rearing of natural enemies for inoculative or inundative release into the cropping environment. Successful candidates from such rearing can be extended to farmers through FFS for adoption. The LA can be validated by research institutions using simple ways of rearing FAW and its natural enemies before adoption. However, the success of mass rearing and release by research institutions require commitment from governments through resource support for sustainable FAW control.

### *5.6. Pheromone trapping and pest monitoring*

Literature shows that, historically, pheromone trap data have been associated with early warning and decision-making with regards to pest control (Silvain 1986; Cruz et al. 2012). Records of trap catches are used for effective guidance in early warning systems for alerting farmers on the intensity of pest scouting needed in fields so that timely control is instituted (Prasanna et al. 2018). However, there are challenges of pheromone specificity as a number of non-target species are being trapped as well. This was confirmed by other authors (e.g. Malo et al. 1999, 2004; Fite et al. 2020).

Two types of traps are currently widely used for FAW monitoring in Africa: the bucket trap with a yellow funnel coupled with insecticide block or liquid soap (Figure 3A) and the white triangular Delta trap fitted with a sticky base (Figure 3B). These traps use synthetic pheromones as lures. Using Delta-type traps with a commercial sex pheromone, Cruz et al. (2012) concluded that moth catches provided the best means for deciding when to apply insecticides for controlling FAW. However, challenges are apparent when lures record zero catches when there is evident infestation by FAW in surrounding fields. Thus, further studies are needed to determine if there is a correlation between FAW trap catches and field infestations. Trapping programmes for African Armyworm [*Spodoptera exempta* (Walker)] have been effective in predicting possible outbreaks in Ethiopia, Kenya, Malawi, Tanzania and Zimbabwe (Negussie et al. 2010). Currently FAO is pioneering a FAW Monitoring and Early Warning System (FAMEWS) for regional early warning and coordinated management of the pest (Ruzzeddu et al. 2018; FAO 2018b; 2019; FAO and CABI 2019). FAMEWS is a mobile application tool that uses data from FAW pheromone trap catches and field scouting (Ruzzeddu et al. 2018) for early warning and scheduling of management operations. The FAMEWS application also provides guiding notes on FAW, other cereal lepidopteran pests, and their natural enemies. However, for the system to be more effective, it requires that pheromones used in trapping be specific so that only adult FAW males are attracted. As has already been alluded to earlier, the lack of correlation between trap catch data and field infestation level requires further study.

Synthetic FAW pheromones are used in monitoring or mass trapping. Examples include PheroLure (active ingredients are a mixture of Z-11-Hexadecen-1-yl Acetate 0.355 mg/lure and Z-9-Tetradecen-1-yl Acetate 1,636 mg/lure, Z-7-Dodecen-1-yl Acetate 0,009 mg/lure) (Insect Science 2020) and Pherodis [active ingredients are a combination of (Z)- 9-tetradecenyl acetate (z-9-14:Ac, (Z)-7-dodecenyl acetate (Z7–12:Ac) and (Z)-9-dodecenyl acetate (Z9–12:Ac)] (Nwanze et al. 2021). However, there is no current evidence comparing the effectiveness of one FAW lure trap type over the other in SSA.

In addition to using pheromone traps and actual field scouting, monitoring of pests in the field can be done using drone technology. This generates reports on the size and spatial distribution of areas infested and can thus be targeted in spray programmes (Filho et al. 2020). This technology is ideal for large tracts of land where walking in fields is time-consuming and practically impossible. Investing in such technology

A

B

**Figure 3.** FAW pheromone trap types: A. Bucket funnel trap (www.greatlakesipm.com); B. Delta trap (Photograph by TH Maulana).

saves on time for scouting and enables early identification of infested spots in the field thus leading to efficient control of the pest. However, application of the drone technology for pest control in Africa is still at its infancy. The technology also has significant initial investment (capital and skills develoipment) unless hired from service providers.

## 6. Policy and regulatory issues in sustainable FAW management

The way in which a pest is controlled can be influenced by national policy (Day et al. 2017). Mechanisms should be put in place in the development of any IPM strategy to ensure that it accommodates measures that are short- and long-term, affordable to farmers and sustainable (Duggal and Siddiqi 2008). Any favourable policy should adequately address protection of biodiversity and human health but funding and lack of expertise are some of the implementation challenges faced (SANBI 2011). In many countries, the acceptance or adoption of strategies considered effective for FAW management have been delayed owing to policy requirements. Such examples include commercialisation of Bt maize in Kenya and Uganda (Norenius 2018) and Argentina (ISAAA 2016b). There are growing demands for GM technological solutions in African countries with Sudan, South Africa and eSwatini having commercialised production while 13 countries are conducting research and testing of GM crops including Nigeria, Ethiopia, Kenya and Malawi (ISAAA 2019). When policies on GM technologies are widely adopted, this may bring another dimension to the sustainable management of FAW to address farmers' challenges.

In Zimbabwe, proposals have been put forward to guide policy on management of FAW with Chimweta et al. (2019) suggesting enforcement of destruction of alternative hosts of FAW during the off-season to break FAW cycles. The growing of maize throughout the year offers hosts for year-round development of FAW. Green mealies that are grown under irrigation for consumption and sale in winter are an important source of income for many smallholder farmers as well as urban vendors. Abandoning off-season growing of green mealies through legislated maize "dead periods" could be a way of breaking FAW life cycle such as is done in the control of pink bollworm, *Pectinophora gossypiella* (Saunders) in cotton and *Myzus persicae* (Sulzer) in tobacco (Chimweta et al. 2019). However, for reasons already mentioned, this is unlikely to be enforceable.

Other areas of policy formulation include review of duty on pesticides and equipment (traps and lures), PPE, reduced registration periods for alternatives to conventional chemical pesticides (e.g. botanicals and biopesticides), subsidies for pesticides used in FAW control and pesticide distribution to the most vulnerable farmers. An example has been Ghana and Zambia's subsidies on products recommended for FAW control thus making the products affordable to smallholder farmers (Rwomushana et al. 2018).

A downward review of duty/taxes for equipment used in production of biopesticides is expected to lower costs of maize production while ensuring that farmers can afford to buy PPE for use in their pesticide spraying operations. Reducing the registration period for alternative pesticides like botanicals would ensure availability of many IPM options for farmers.

The presence of FAW on a number of hosts has a bearing on a country's potential to trade. FAW is regarded as a pest of quarantine importance by the EU with strict phytosanitary requirements on trade to prevent its introduction into Europe (Jeger et al. 2017). *Dendranthema* spp. and Cucurbitaceae from Ghana, *Dianthus* spp. and *Solanum* spp. from South Africa and Zimbabwe have been documented as potential pathways for introduction of FAW into the EU (Jeger et al. 2017). Strengthening the capacity to inspect and detect the pest at ports of exit would secure trading opportunities for most countries.

## 7. Awareness-raising, farmer training and ICT

Farmer education and communication are important elements under FAO's FAW management interventions (Feldmann et al. 2019) and have been key to its correct identification and management. Pest awareness and early response contribute to reduced pest damage (Kansiime et al. 2019). In Uganda, Tambo et al. (2019) identified limited awareness and knowledge of FAW as having adverse effects on its management. A number of governments embarked on nation-wide training programmes to raise awareness on the new pest and to capacitate farming communities and stakeholders with resources to manage it. Subject areas covered in the awareness and training campaigns included FAW biology, identification, ecology, management and safe use of pesticides; thus addressing some of the challenges highlighted by Chimweta et al. (2019). Regional and international organisations such as CABI, FAO, CIMMYT, ICIPE and IITA are contributing towards development of training and information materials and mobile application tools for use by farmers and other stakeholders. Demonstrations on field scouting and application of pesticides are also part of the training curricula.

Field evidence shows that agro-dealers are developing their own mobile tools to enhance farmer's knowledge and avail information on products that are useful in addressing their challenges and locations of nearby outlets selling agrochemicals. Use of electronic media also plays a significant role as an ICT-based campaign strategy (Tambo et al. 2019). However, the costs of such gadgets may be beyond the reach of many and in cases where data and internet connectivity are a requirement, challenges are compounded (Ruzzeddu et al. 2018).

Several Information Communication Technologies (ICT) have been brought forward for use by farmers, extension staff, policy makers, Regional and International Plant Protection Organisations, and researchers, among a variety of users to assist in decision-making and well-informed management of FAW in different parts of the world (Table 3). However, with such technologies other challenges come that relate to cost and access as some rely on data bundles or internet service charges. Feldmann et al. (2019) indicated that FAMEWS does not provide easy-to-handle recommendations for proper management of FAW. The incorporation of artificial intelligence Nuru - an innovative talking app that uses cutting-edge technologies involving machine learning and artificial intelligence - into FAMEWS has enabled farmers to recognize FAW and take immediate measures to destroy it (Otim et al. 2021). There is great potential for further development and adoption of these digital tools and approaches in many African countries towards efforts to mitigate the impact of FAW. Valuable insights could be derived from independent comparative evaluation of the available ICT solutions under African conditions.

## 8. Adoption of FAW management technologies

The negative impact of FAW on crop production has resulted in the adoption of management strategies that are effective and readily available to smallholder farmers in most countries of SSA. However, access to knowledge of both the pest and the control strategies have played a critical role in the selection and adoption of best control strategies. Rwomushana et al. (2018) reported relatively low adoption of effective strategies for FAW control such as "push-pull" while handpicking egg masses and caterpillars were popular in SSA. Factors contributing to low adoption include, among others, availability of seed, labour to establish companion crops and adaptation to farming systems (Rwomushana et al. 2018). The decision to adopt a FAW management strategy varies with farming system, i.e. commercial or smallholder farming systems. In most commercial production systems, strategies adopted for FAW management include chemical control while smallholders farmers utilise chemical control, handpicking and crushing, crop rotations, application of sand, ashes, and detergents, among others. In some cases, households may not adopt any strategy to control FAW (non-adopters). In their findings, Tambo et al. (2019) indicated that the adoption of a FAW management strategy was associated with education level attained within households.

The role played by governments at policy level has an influence on adoption of management strategies, especially when there is promotion of use of biopesticides, low subsidies to support their production and IPM policies. The uptake of biopesticides in the fight against FAW is still low due to their viability challenges in the environment, especially under low precipitation and extreme temperatures, unknown storage conditions, and the need for special application equipment (Gasic and Tanovic 2013; Bateman et al. 2018).

Citizen science is taking centre-stage in involving farmers in the research process, through data collection and interpretation of results, for ease of adoption of technologies. Citizen science involves nonscientists, in this case farmers, working with scientists in the process, methods and standards of research with the intention of advancing scientific knowledge and application (National Academies of Sciences et al. 2018). Evidence of farmers experimenting with other traditional solutions to fight FAW exists for example the use of ash and detergents to control FAW larvae in Ghana, Zambia and Zimbabwe (Tambo et al. 2019, 2021; Asare-Nuamah 2022). In Ethipoia, Kumela et al. (2019) also provided evidence of farmers widely using traditional methods for FAW control. A number of non- and quasi governmental organisations are now involved in promoting citizen science including IITA on FAW control (Feldmann et al. 2019), and ICRISAT on management of groundnut pests in India (Wightman 2018).

In Zimbabwe, farmers are involved in "push-pull" research (Baudron et al. 2019) and this provides evidence through their involvement in coming up with effective technologies against FAW. Participatory research through FFS involves nonscientists being involved in coming up with ways of solving their challenges. FAO has promoted FFS as platforms for farmers to learn and exchange information. To date, more than 90 countries are using the FFS approach. A key policy priority by governments should therefore be to allocate land designated for FFS and support extension staff in their operations for long-term sustainable farmer and community education. A Global Farmer Field School platform has also been created

**Table 3.** Regional and International ICT platforms developed for use by farmers, extension staff and other stakeholders for fall armyworm monitoring and management.

| Country/Region | ICT technology | Brief description of function | Source |
|---|---|---|---|
| South Africa | Africa Rising | An SMS-based interactive service assisting farmers to prevent and mitigate FAW by acting appropriately to respond based on the severity, likely impact on the current crop, and cost-benefit. | Rwomushana et al. (2018) |
| Israel | AI-based Digital Monitoring System | A digital monitoring system enabled by an app that leverages and deploys artificial intelligence (AI) to provide diagnoses and generate warning alerts for FAW outbreaks based on geographical spread and forecasting models. | |
| Ghana | Boa Me | An artificial intelligence-powered, web-based application that combines past and satellite data, and user information about FAW and translates it into insights that farmers can access through local voice systems, SMS alerts, or a public announcement system. | |
| | CdPAS (Crop Disease & Pest Advisory Service) | This combination of an interactive voice response and smart app allows farmers to report, predict, identify monitor and mitigate outbreak of FAW across Africa. It also provides crop disease prediction and advisory services, in addition to providing education and awareness through interactive voice response messaging in local languages. | |
| | Igeza | A cloud-based mobile app enabling early detection of FAW via video, image or voice, with geolocation, which then sends all notifications to a pool of experts (entomologists, pathologists, agronomists etc.) to analyse the data and recommend appropriate control options. | |
| Nigeria | CornBot | An audio-visual mobile maize disease diagnostic app that interacts with farmers in their local languages to help them identify, track, map, and treat FAW. | |
| | FarmSmart FAW CONTROL | An integrated image recognition app for FAW that links solution providers (researchers, manufacturers, and financial institutions) with farmers and allows paying a fraction of the control cost and rating the effectiveness of the solution provided. | |
| Taiwan | Digicult | A digital platform using mobile communication devices, multimedia processing, and crowdsourcing to analyse and share FAW data, and provide advice which helps farmers prevent and fight FAW. | |
| Uganda | EzyAgric | A pest and disease diagnostic app, utilizing the EzyArmyWorm (EAW) module that allows farmers to detect FAW life stages, and predict yield loss. | |
| | Agri-Poll | Smart survey platform that allows extension workers and key stakeholders to gather, analyse, and disseminate images (of pests and crops), location, time, weather, and environments in which the pests have been detected via feature phones, smart phones, and web platform. | |
| | FAWLEA: Pests? Problem solved! | A mobile phone app that uses image recognition to detect and accurately identify FAW and the extent of crop damage, then provide the farmer with control options and approved pesticides and agro-input suppliers. | |
| United Kingdom | Fall Armyworm Virtual Advisor | A combination of a chatbot and a progressive web app, that guides farmers through a training on how to identify and manage FAW. | |
| United States of America | myAgro | A mobile app that allows myAgro field agents to easily report and act on FAW prevalence based on heat maps, as they monitor farmers' field. myAgro provides a digital payment platform enabling farmers to use their mobile phones to pay incrementally for FAW specific training, seeds, and other products, thus helping farmers to prepare and better deal with FAW attacks on their crops. | |
| | Light Watch: Autonomous Pest Monitoring Solution | A monitoring solution using solar-powered LED lure, intelligent cameras, and a convolutional neural network to detect, alert central pest management authorities, and local farmers via SMS of FAW outbreaks. | |
| | UDefeatFAW | Interactive radio and SMS/interactive voice response for two-way communication with smallholder farmers for delivery of vouchers for appropriate and affordable pesticides, and the correct stage of application and provides guidance on control of FAW. | |
| | WeFAW | An SMS-based FAW early warning, response and control model for smallholder farmers utilizing data from millions of farmer interactions on "Wefarm" platform to provide real-time 'alerts' and access to the knowledge and resources from subscribed smallholder farmers. | |
| | AfriFARM – Africa Fall Armyworm Response Mechanism | Mobile phone app that provides current, quality information about FAW tailored to meet user needs by overcoming technology gaps of connectivity and data/SMS costs, as well as literacy gaps. The app is designed to facilitate education, monitoring, and safe and effective responses to the threat of FAW. | |
| Kenya | Shape Up Against Armyworm | A make-over reality TV and radio, print, mobile and internet accessible programme that provides up-to-date information weekly to farmers through the Shamba Shape Up TV/Radio series on FAW control, access to inputs, and links to further information, via the iShamba mobile call centre and SMS services. | |
| | Lindafarm | An app that employs mobile technology, and databases/maps, to help farmers send alerts on FAW on their farms via USSD/SMS and get instant help from nearby community extension service providers. | |
| | Zaois-Tech | Allows farmers and service providers to use one or more of the system digital tools to identify, eliminate, and prevent FAW using either an SMS text-based query system, a smartphone app, a website portal and/or an image identification tool. | |
| International/ Global | FAMEWS-FAO | A mobile app for collecting, recording and transmitting scouting and trap data, IPM education, digital library, chat platform to share experiences and provide expert resources. | Ruzzeddu et al. (2018); FAO (2018); Feldmann et al. (2019); FAO and CABI (2019) |
| | FAWRisk Index-FAO | Models the risk of food insecurity due to a FAW outbreak. | Ruzzeddu et al. (2018) |
| | FAW impact assessment-FAO | Data collected using KoBo toolbox which is a digitalised questionnaire and analysis system accessed through smartphones for a better understanding of the impacts of FAW on crop yield, particularly maize crop yield as well as the impact of control measures themselves. | Ruzzeddu et al. (2018); FAO (2018) |
| | FAW dashboard tool-FAO | A tool to monitor the prevalence of FAW across each country and published at least once per agricultural season to give decision makers a high-level overview of the prevalence of FAW, populations affected, and the current status of FAW-related response programmes. | Ruzzeddu et al. (2018) |
| **International/ Global** | Plantwise Factsheet App - CABI | Provides a library of clear, practical and safe advice for tackling crop problems. | www.cabi.org/isc/fallarmyworm. . . . . . |

by FAO as a knowledge centre on FFS (FAO 2018c). Through these FFS, farmers have been involved in the use of available and cheap alternatives to managing FAW, and training on identification and conservation of natural enemies of FAW. It is through these FFS that augmentation of FAW natural enemies can be strengthened. FFS provide hands-on practical demonstrations run by farmers allowing for farmer-based observations and recommendations for working solutions to FAW management. This in turn allows for easy adoption of working strategies against FAW. Some of the strategies that have been tested by farmers in FFS include use of "push-pull" strategy, natural plant extracts, conservation agriculture, and early planting, among a range of options in FAW management. In this case, farmers decide on which measures to experiment on with guidance from extension staff. Other benefits of FFS have been the inclusion of nutrition crops like bio-fortified sweet potatoes, beans and maize as well as training on agroecosystem analysis, record keeping and IPM.

## 9. Integration of approaches

Any IPM strategy requires expert support to farmers for appropriate decision-making and involves a number of steps (Figure 4). Several combined strategies are already being implemented in the management of FAW in SSA. In Zambia and Zimbabwe, for example, a notable combined strategy is that of the seed treatment chemical Fortenza$^{TM}$ Duo and other chemical insecticide sprays. The strategy has merits in that farmers do not have to apply insecticides for the first 3–4 weeks of growth, allowing them time to carry out other operations, reduces health risks from exposure to pesticide applications, reduces environmental contamination, while promoting natural enemy build-up, especially predators. Use of treated seed is now a management technology choice available for farmers. Similarly, the use of other approaches such as resistant or tolerant crop varieties in an IPM strategy would be beneficial (Matova et al. 2020).

The inclusion of biocontrol as an important component in IPM requires extensive surveillance and documentation of FAW natural enemies followed by raising awareness on their availability, mass rearing and release programmes, pest monitoring and correct decision-making on when to control based on economic threshold levels (Singh and Gupta 2017). With existing information of FAW distribution and levels of damage it causes in maize growing areas in Africa (Table 1), timely decisions can be made on when to apply chemicals based on scouting information and thresholds.

However, the choice of strategies to use may be affected when policy promotes the importation and use of synthetic pesticides. The promotion of bio-pesticides provides compatible products that have less impact on natural enemies of FAW.

In countries were "push-pull" strategy is being employed for FAW management, the strategy is combined with either conservation agriculture or early planting followed by Good Agronomic Practices to reduce the impact of FAW with farmers getting high yields. The strategy has other benefits of improved nutrition from legumes, improved stockfeed, striga control, increased FAW natural enemy activity and increased income from high yields, and increased nitrogen and soil humidity (Prasanna et al. 2018; FAO 2018c). Countries where farmers have adopted "push-pull" strategies with positive effects on yields include, among others, Kenya, Uganda, Tanzania (Midega et al. 2018) and Zimbabwe (Baudron et al. 2019).

Existing literature shows no ideal IPM package that has been developed and tested for universal use by farmers. Most research work recognises the importance of IPM with recommendations for specific management technologies to be included in the strategy. However, at farmer level, there is need to refine context-specific packages that consider available resources, existing natural enemy complex, social acceptance (Dara 2019; Otim et al. 2021), labour intensiveness (Chimweta et al. 2019; Kansiime

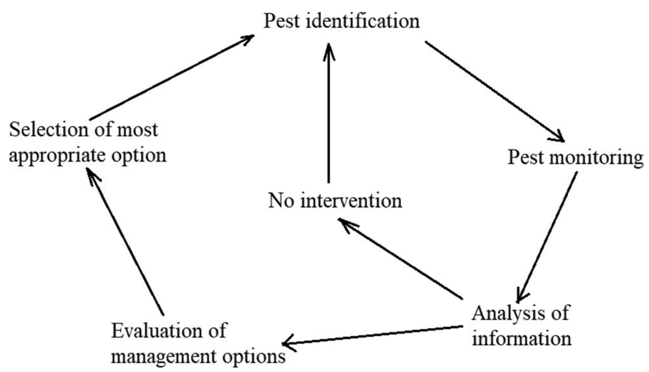


Figure 4. Decision making process in IPM (Adapted from Singh and Gupta 2017).

et al. 2019; Tambo et al. 2019, 2020, 2021), sustainability (FAO 2018b), and risks associated with the strategy involved (Duggal and Siddiqi 2008), among other factors. Different regions are bound to adopt different combinations of strategies based on their experiences and scientists should work with farmers in providing evidence-based IPM strategies that are effective in controlling FAW in specific contexts.

At national, regional and international level, governments, research institutions and other organisations may contribute to development and use of effective IPM packages by enhancing a policy, regulations and research environment conducive for collaboration with farmers to ensure adoption of sustainable and effective area-wide FAW management packages ideal for both commercial and subsistence farmers.

## 10. Conclusions

In just less than five years after it was first recorded on the African continent, FAW has become the most economically damaging pest of vegetative-stage maize. Attacks on maize are a threat to food and nutrition security, particularly in regions where maize is a staple food crop. Several techniques are currently being employed in the management of FAW. However, the techniques require an assessment of the best compatible options for inclusion in an IPM-based sustainable management strategy. The existence of common natural enemies across several African countries can be exploited to enhance biocontrol of FAW. This can be achieved though conservation, inoculative or inundative releases of the most efficient parasitoids. Complementary strategies such as “push-pull” would contribute to reduced costs of FAW control for smallholder farmers, especially under conditions of reduced rainfall and global warming. Challenges on the adoption of GM crops by most African countries require further research, along with awareness-creation to farmers, extension agents and policy makers. An IPM approach is the mainstay for sustainable FAW management, but needs well-informed policies to harness synergistic effects of various management strategies. Effective monitoring of FAW requires an efficient trapping system plus field scouting to provide early warning data, which will guide where to direct resources for best FAW management. While resistance development has been reported in Bt maize hybrids, measures to slow it down coupled with alternative strategies such as biological, cultural and “push-pull” strategies may offer lasting solutions to FAW management. With the development in technology, ICT comes in handy in improving pest management through awareness-creation, information sharing among farmer groups or association, scouting and monitoring as well as communicating alerts for effective pest control. Ultimately, for sustainable uptake of identified FAW management strategies, socioeconomic circumstances of users need to be studied. The environment under which sustainable pest management can also be achieved requires supporting policies and regulations that ensure that affordable and low-risk technologies are readily available to farmers.

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## ORCID

Brighton M. Mvumi http://orcid.org/0000-0002-3822-6545

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